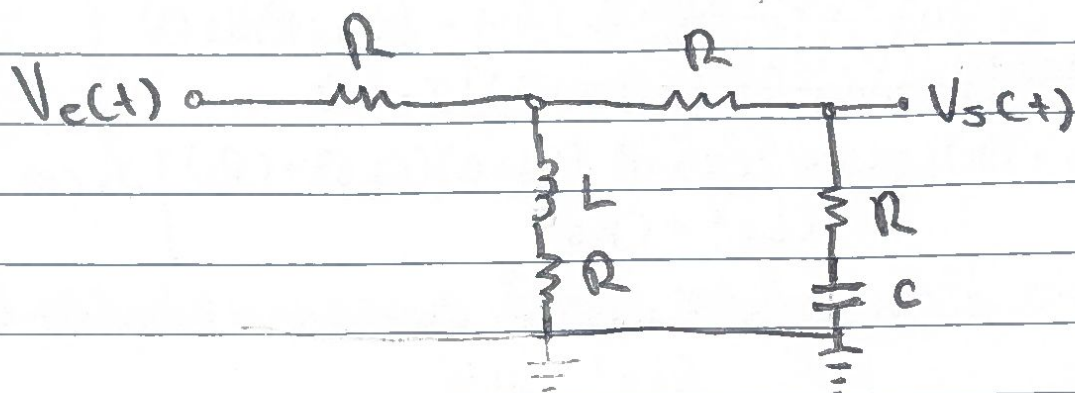


Práctica: Diseño de controlador para un sistema de segundo orden



Ec. principales ~~~

$$V_e(t) = R\dot{i}_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace ~~~

$$V_e(s) = RI_1(s) + Ls(I_1(s) - I_2(s)) + R[I_1(s) - I_2(s)]$$

$$Ls[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)] = RI_2(s) + RI_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = RI_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico ~~~

- 1.- Agrupar corrientes $I_1(s)$ e $I_2(s)$
- 2.- Expresar el $V_e(s)$ en función de $I_2(s)$
- 3.- Despejar $I_1(s)$ de la segunda ec. principal.

$$V_s(s) = \left(R + \frac{1}{Cs}\right) I_2(s) = \left(\frac{CRs + 1}{Cs}\right) I_2(s)$$

$$LsI_1(s) - LsI_2(s) + RI_1(s) - RI_2(s) = RI_2(s) + RI_2(s) + \frac{I_2(s)}{Cs}$$

$$LsI_1(s) + RI_1(s) = RI_2(s) + RI_2(s) + \frac{I_2(s)}{Cs} + LsI_2(s) + RI_2(s)$$

$$(Ls + R)I_1(s) = \left(Ls + \frac{1}{Cs} + 3R\right) I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R\right) I_2(s)$$

$$I_1(s) = \frac{CLs^2 + 3(CRs + 1)}{CLs^2 + CRs} I_2(s)$$

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$$V_e(s) = R I_1(s) + L s I_1(s) - L s I_2(s) + R I_1(s) - R I_2(s)$$

$$V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$V_e(s) = (Ls + 2R) \frac{CLs^2 + 3(Rs + 1)}{CLs^2 + CRs} I_2(s) - (Ls + R) I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3(Rs + 1))}{CLs^2 + CRs} - (Ls + R) \right] I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3(Rs + 1)) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs} \right] I_2(s)$$

$$V_e(s) = \left[\frac{CL^2s^3 + 3LCRs^2 + Ls + 2LCRs^2 + 6CR^2s + 2R - (CL^2s^3 + LCRs^2 + CLRs^2 + CR^2s)}{CLs^2 + CRs} \right] I_2(s)$$

$$V_e(s) = \left[\frac{3LCRs^2 + 5CR^2s + 2R + Ls}{CLs^2 + CRs} \right] I_2(s)$$

$$\frac{V(s)}{V_e(s)} = \frac{\frac{(Rs + 1)}{Cs} I_2(s)}{\frac{3LCRs^2 + 5CR^2s + 2R + Ls}{CLs^2 + CRs} I_2(s)}$$

$$= \frac{\frac{(Rs + 1)}{Cs} I_2(s)}{\frac{3LCRs^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)} = \frac{((Rs + 1)(Ls + R))}{3CLRs^2 + (L + 5CR^2)s + 2R}$$

Error en estado estacionario cuando se aplica un escalón unitario de entrada.

$$e(s) = \lim_{s \rightarrow 0} \frac{1}{s} \left(1 - \frac{CLRs^2 + (CR^2 + L)s + R}{3CLRs^2 + (5CR^2 + L)s + 2R} \right) = \frac{R}{2R}$$

$$e(s) = 0.5V$$

Estabilidad en lazo abierto

Se calculan los polos del sistema (raíces del denominador)

$$R = \Omega ; L = H ; C = F \quad R = 10k\Omega ; L = 10mH ; C = 220\mu F$$

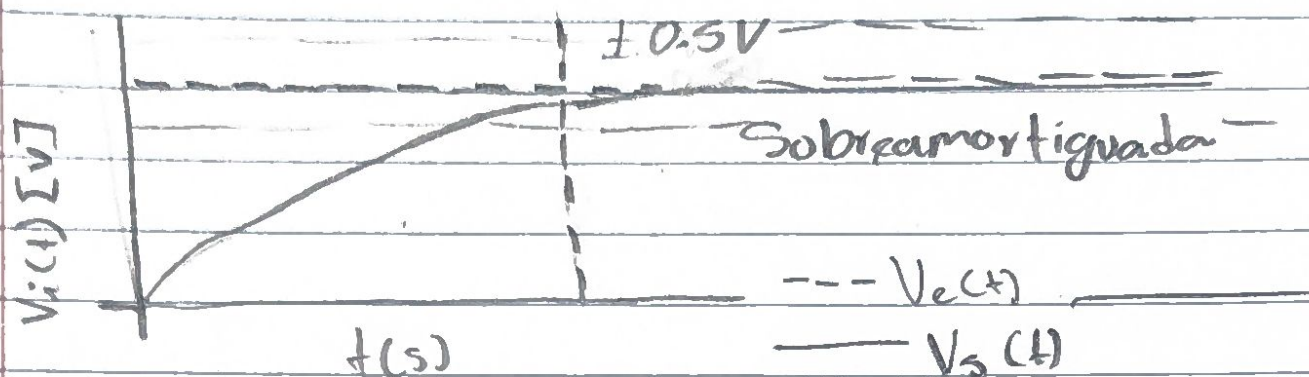
$$3CLRs^2 + (5CR^2 + L)s + 2R = 0$$

$$0.066s^2 + 110,000s + 20k = 0$$

$$\lambda_1 = -0.1818 \quad \lambda_2 = -1,666,666.636$$

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Bosquejo



Modelo de ecuaciones integro-diferenciales

A partir de las ecuaciones principales del sistema, se despejan las variables dependientes, es decir, las corrientes y el voltaje de salida:

$$i_1(t) = [V_e(t) + R i_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}] \frac{1}{2R}$$

$$V_e(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = [L \frac{d[i_1(t) - i_2(t)]}{dt} + R i_1(t) - \frac{1}{C} \int i_2(t) dt] \frac{1}{3R}$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$R = 10k\Omega \quad L = 10mH \quad C = 220\mu F$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLRs^2 + (CR^2 + L) + R}{3CLR s^2 + (5(R^2 + L)s + 2R)} = \frac{0.022s^2 + 22000.01s + 10000}{0.066s^2 + 110000.01s + 20000}$$